

AI Assistant lab-21.3

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Batch:41

TASK 1 – AI-Assisted Expense Tracker

Initial prompt:

Create a simple Python expense tracker to store daily expenses with category and amount.

Initial code:

```
expenses = []

while True:
    name = input("Enter expense name: ")
    amount = float(input("Enter amount: "))
    category = input("Enter category: ")

    expenses.append((name, amount, category))

    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

print("Expenses:", expenses)
```

Output:

```
Enter expense name: food
Enter amount: 100
Enter category: groceries
Add more? (y/n): y
Enter expense name: transport
Enter amount: 50
Enter category: travel
Add more? (y/n): n
Expenses: [('food', 100.0, 'groceries'), ('transport', 50.0, 'travel')]
```

Improvement prompt:

Improve the expense tracker with monthly summary, input validation, and functions.

Improved code

```
def add_expense(expenses):
```

```

name = input("Enter expense name: ")

while True:
    try:
        amount = float(input("Enter amount: "))
        break
    except:
        print("Invalid amount")

category = input("Enter category: ")
expenses.append({"name": name, "amount": amount, "category": category})

def show_summary(expenses):
    total = sum(e["amount"] for e in expenses)
    print("Total Monthly Expense:", total)

expenses = []

while True:
    add_expense(expenses)
    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

show_summary(expenses)

```

Output:

```

/home/shashidhar/Downloads/aiassitantcoding-20260312T084629Z-3-001/aiassitantcoding/65.py
Enter expense name: food
Enter amount: 100
Enter category: groceries
Add more? (y/n): y
Enter expense name: transport
Enter amount: 50
Enter category: travel
Add more? (y/n): n
Total Monthly Expense: 150.0

```

Justification

- Functions → modular design

- Validation → avoids runtime errors
- Summary → real-world usability

TASK 2 – Git Version Control (Student Grade Calculator)

Initialize Repo,Initial Code,Create Branch

Ai prompt

Add grade classification and percentage calculation to the program.

Updated

```
>
> total = sum(marks)
> avg = total / len(marks)
> percentage = (total / (len(marks)*100)) * 100
>
> if avg >= 90:
>     grade = 'A'
> elif avg >= 75:
>     grade = 'B'
> elif avg >= 50:
>     grade = 'C'
> else:
>     grade = 'F'
>
> print("Average:", avg)
> print("Percentage:", percentage)
> print("Grade:", grade)
> EOF
Switched to a new branch 'feature-ai-grade'
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$ git add .
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$ git commit -m "Added grade classification and percentage calculation"
marks = list(map(int, input("Enter marks: ").split()))

total = sum(marks)
avg = total / len(marks)
percentage = (total / (len(marks)*100)) * 100

if avg >= 90:
    grade = 'A'
elif avg >= 75:
    grade = 'B'
```

```

elif avg >= 50:
    grade = 'C'
else:
    grade = 'F'

print("Average:", avg)
print("Percentage:", percentage)
print("Grade:", grade)

```

Commit + Merge

Justification

```

--- grade.py ---
marks = list(map(int, input("Enter marks: ").split()))

total = sum(marks)
avg = total / len(marks)
percentage = (total / (len(marks)*100)) * 100

if avg >= 90:
    grade = 'A'
elif avg >= 75:
    grade = 'B'
elif avg >= 50:
    grade = 'C'
else:
    grade = 'F'

print("Average:", avg)
print("Percentage:", percentage)
print("Grade:", grade)
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$ git branch -m master main && git checkout main && git merge feature-ai-grade && git log --oneline
e --graph --all --max-count=5
Switched to branch 'main'
Updating 15776bb..649c76d
Fast-forward
 grade.py | 17 ++++++++
 1 file changed, 16 insertions(+), 1 deletion(-)
* 649c76d (HEAD -> main, feature-ai-grade) Added grade classification and percentage calculation
* 15776bb Initial grade calculator with average

```

- Branching → safe feature development
- Merge → integration workflow
- AI → faster feature addition

TASK 3 – AI Commit Messages (Password Validation)

Setup code

```

import re

password = input("Enter password: ")

if len(password) < 8:

    print("Weak password")

elif not re.search(r'[@#$$%^&*]', password):

    print("Add special character")

else:

    print("Strong password")

```

Prompt

Generate commit message for adding password validation with length and special character rules.

AI Commit Message

```

fatal: not a git repository (or any of the parent directories): .git
shashidhar@shashidhar-750XGK:~/Downloads/aissitantcoding-20260312T084629Z-3-001$ git commit -m "feat: add password validation with minimum length and special character requirement"

```

Human Version

Added strong password validation rules

Justification

- AI → more structured (uses “feat:” convention)
- Human → less descriptive

TASK 4 – Quiz Management System (Pair Programming)

Student A Code

```
questions = {
    "Capital of India?": "Delhi",
    "2+2?": "4"
}

for q in questions:
    ans = input(q + " ")
    if ans == questions[q]:
        print("Correct")
    else:
        print("Wrong")
```

AI Prompt (Student B)

Add score calculation and input validation to quiz program.

Final code

```
questions = {
    "Capital of India?": "Delhi",
    "2+2?": "4"
}

score = 0

for q in questions:
    ans = input(q + " ").strip()

    if ans.lower() == questions[q].lower():
        print("Correct")
        score += 1
    else:
        print("Wrong")

print("Final Score:", score)
```

Git Workflow

```

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git checkout -b feature-score
Switched to a new branch 'feature-score'

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git add .

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git commit -m "Added score calculation"
[feature-score 8ab91cd] Added score calculation
1 file changed, 12 insertions(+), 1 deletion(-)

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git checkout main
Switched to branch 'main'

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git merge feature-score
Updating 2f4a7d1..8ab91cd
Fast-forward
 quiz.py | 12 ++++++++--
1 file changed, 11 insertions(+), 1 deletion(-)

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ python quiz.py
Capital of India? Delhi
Correct
2+2? 5
Wrong
Final Score: 1

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git log --oneline --decorate --graph --all --max-count=5
* 8ab91cd (HEAD -> main, feature-score) Added score calculation
* 2f4a7d1 Initial quiz management system

```

Output:

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS SPELL CHECKER 69 SONARQUBE 4 Python - Downloads + v [ ] [ ] ...
shashidhar@shashidhar-750XGK:~/Downloads$ /usr/bin/python3 /home/shashidhar/Downloads/aiassitantcoding-20260312T084629Z-3-001/aiassitantcoding/65.py
Capital of India? y
Wrong
2+2? 4
Correct
Final Score: 1
shashidhar@shashidhar-750XGK:~/Downloads$

```

Justification

- Score → measurable output
- Validation → better UX
- Collaboration → real Git workflow

